Games with sequential actions (perfect/imperfect information)

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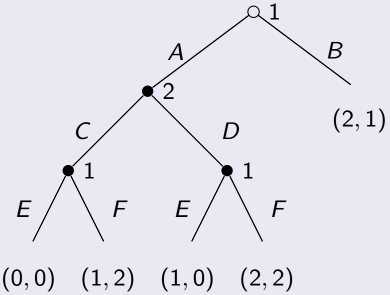
Extensive-form games: definitions

Definition

The ***extensive-form representation*** of a game is a game tree in which nodes represent players' choices, edges represent the actions available to the players and leaves the final outcomes, with their utility values.

Example



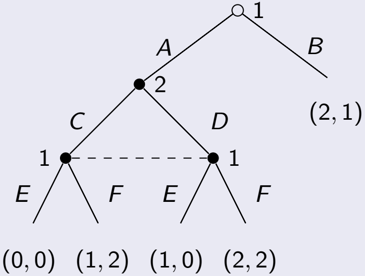
Extensive-form games: definitions

Remark

Players can have imperfect knowledge of the past actions (*imperfect information*) in extensive-form games.

Example

Suppose that player 1 does not distinguish between C and D.



Extensive-form games: definitions

A ***history*** $h = (a^1, \ldots, a^k)$ is the unique path of actions from the root to a node h of the game tree.

Information set: a set I_i (dashed line) containing two or more non-terminal nodes at which a player i does not distinguish between previous histories.

The collection $\mathcal{I}_i = (I_{i,1}, \ldots, I_{i,k_i})$ is the ***information partition*** of i .

An ***extensive-form game of imperfect information*** is a game in which at least one information set is not a singleton.

Example

In the previous example:
$\mathcal{I}_1 = \{I_{1,1}, I_{1,2}\} = \{\{\emptyset\}, \{(A, C), (A, D)\}\}$, $\mathcal{I}_2 = \{I_{2,1}\} = \{\{A\}\}$.

Extensive-form games: definitions

Definition

A ***pure strategy*** for player i is a function s_i that assigns actions $a \in A$ to every information set available to player i.

Definition

The ***outcome*** $O(s)$ of a strategy profile $s = \{s_i\}_{i \in \mathcal{N}}$ is the terminal node reached when each player $i \in \mathcal{N}$ plays according to s_i.

Example

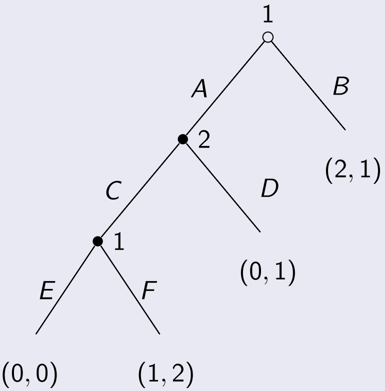
In the example in slide 2: $s_1 \in \{AEE, AEF, AFE, \dots, BEE, \dots\}$, $s_2 \in \{C, D\}$. The outcome $O\left((AEE, C)\right)$ gives payoff profile $(0, 0)$.

In the example in slide 3: $s_1 \in \{AE, AF, BE, BF\}$, $s_2 \in \{C, D\}$. The outcome $O\left((AF, C)\right)$ gives payoff profile $(1, 2)$.

Normal-form representation of extensive-form games

Remark

Every extensive-form game has a normal-form representation (possibly exponential in the tree depth).



	C	D
AE	0, 0	0, 1
AF	1, 2	0, 1
BE	2, 1	2, 1
BF	2, 1	2, 1

	C	D
AE	0, 0	0, 1
AF	1, 2	0, 1
B	2, 1	2, 1

Nash Equilibria in extensive-form games

Definition

A ***(pure strategy) Nash equilibrium*** of an extensive-form game G is a strategy profile s^* such that $\forall i \in \mathcal{N}$

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*), \quad \forall s_i \in S_i.$$

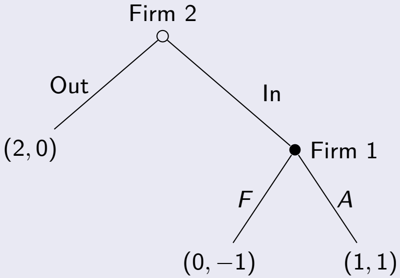
The history h_* generated by s^* is called *equilibrium path*.

Remark

In a Nash equilibrium, each player's extensive-form strategy needs to be optimal only along the equilibrium path.

In the previous example, $((B, *), C)$, $((B, *), D)$ are NEs.

Example (Entry Game)



	Out	In
F	2, 0	0, −1
A	2, 0	1, 1

Nash equilibria: (F, Out), (A, In).

Equilibrium (F, Out) is sustained by the threat that Firm 1 chooses F if Firm 2 enters the market.
Is this a ***credible threat***? No, it is not *sequentially rational* for Firm 1.

Subgame-perfect equilibria

The *subgame* of G rooted at node h is the restriction of G to the descendants of h.

Definition

A strategy profile s is a ***subgame-perfect equilibrium*** (SPE) of an extensive-form game (with perfect information) G if for any subgame G' of G, the restriction of s to G' is a Nash equilibrium of G'.

Theorem (Kuhn, 1953)

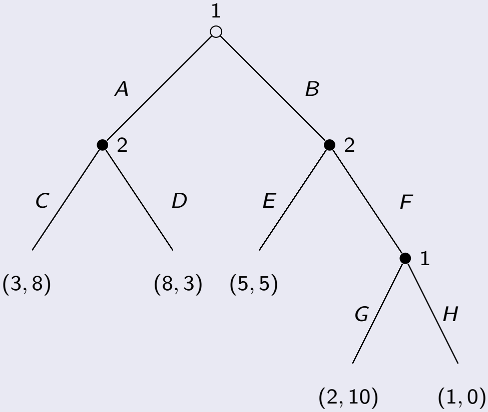
Every finite extensive-form game (with perfect information) has a ***pure-strategy SPE****.*

Remark

A SPE can be found applying the ***backward induction*** process.

Subgame-perfect equilibria

Example



	CE	CF	DE	DF
AG	3, 8	3, 8	8, 3	8, 3
AH	3, 8	3, 8	8, 3	8, 3
BG	5, 5	2, 10	5, 5	2, 10
BH	5, 5	1, 0	5, 5	1, 0

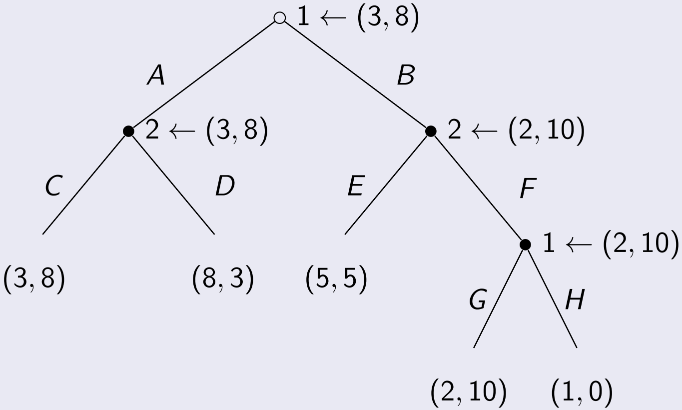
	CE	CF	DE	DF
A	3, 8	$\underline{3, 8}$	8, 3	8, 3
BG	5, 5	2, 10	5, 5	2, 10
BH	$\underline{5, 5}$	1, 0	5, 5	1, 0

Pure-strategy N.E.: $((A, G), (C, F))$, $((A, H), (C, F))$, $((B, H), (C, E))$.

Subgame-perfect equilibria

Example (cont'd)

Backward induction analysis of the game tree: $((A, G), (C, F))$ is SPE.



Behavioral strategies in extensive-form games

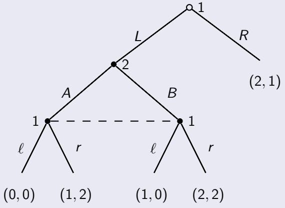
A ***mixed strategy*** is a randomization over pure strategies.

Example

Pure strategies for player 1:
(L, ℓ), (L, r), (R, ℓ), (R, r)

Mixed strategy $\sigma_1 = (p_{11}, p_{12}, p_{13}, p_{14})$

The probability of playing L at $I_{1,1}$ is $p_{11} + p_{12}$.
The probability of playing ℓ at $I_{1,2}$ is $p_{11} + p_{13}$.



Remark

In a mixed strategy, choices at different information sets are *correlated*.
(Difficult to work with!)

Behavioral strategies in extensive-form games

Example

Suppose player 1 randomizes over L and R with probability p and between ℓ and r with q corresponding to the mixed strategy

$$\sigma_1 = (pq, p(1-q), (1-p)q, (1-p)(1-q)) \, .$$

Let us define a "behavioral" strategy $\beta_1 = ((p, 1-p), (q, 1-q))$.

Definition

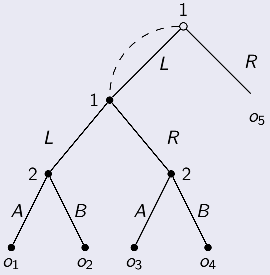
A ***behavioral strategy*** β_i for player i in an extensive-form game is a collection $\{\beta_i [I_{i,j}]\}_{I_{i,j} \in \mathcal{I}_i}$ of independent probability measures $\beta_i [I_{i,j}]$ on $A(I_{i,j})$ defined on all information sets $I_{i,j}$ available to i.

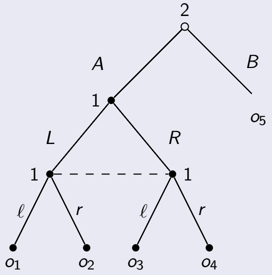
Remark

More compact representation than mixed strategies.

Example

Mixed and behavioral strategies are not always equivalent.





Left: o_3, o_4 unreachable using mixed strategies, with prob $p(1-p)$ using behavioral ones.
Right: No behavioral strategy for $\sigma_1 = \frac{1}{2}(L,\ell) + \frac{1}{2}(R,r)$. [$\nexists p, q$ s.t. $p(1-q) = 0 = (1-p)q$ and $pq = 1/2$.]

Behavioral strategies in extensive-form games

Games that do not present these pathological situations are called of **perfect recall**.

In finite extensive-form games with perfect recall, mixed strategies and behavioral strategies are (outcome) equivalent.

Proposition

A finite extensive-form game in which all players have perfect recall admits a Nash equilibrium in behavioral strategies.

Proof.

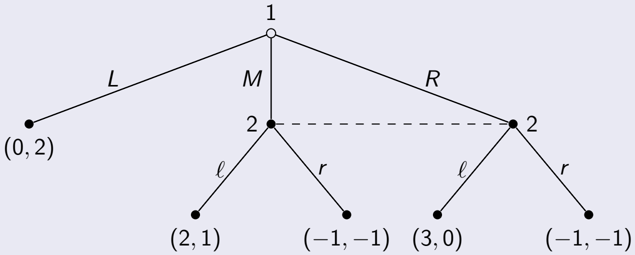
A finite extensive-form game is equivalent to a finite normal-form game. Because of perfect recall, mixed and behavioral strategies are equivalent, thus the result follows from Nash theorem. □

Nash equilibria and perfection

Do Subgame Perfect Equilibria still exist? Yes, with some issues:

- subgames must preserve the information sets
- often there are no proper subgames
- players can threaten to play strictly dominated strategies at off-equilibrium information sets

Example



	ℓ	r
L	$0, 2$	$0, 2$
M	$2, 1$	$-1, -1$
R	$3, 0$	$-1, -1$

Beliefs, rationality and Bayes

Definition

$\forall i \in \mathcal{N}$, a ***belief*** $\mu_i : \mathcal{I}_i \to [0,1]$ is a function that assigns to every information set $I_{i,j} \in \mathcal{I}_i$ a probability measure, i.e. $\mu_i[I_{i,j}](h) \geq 0$, $\forall h \in I_{i,j}$ and $\sum_{h \in I_{i,j}} \mu_i[I_{i,j}](h) = 1$.

Definition

An ***assessment*** (β, μ) is a pair in which $\beta = \{\beta_i\}_{i \in \mathcal{N}}$ is a behavioral strategy profile and $\mu = \{\mu_i\}_{i \in \mathcal{N}}$ is a system of beliefs.

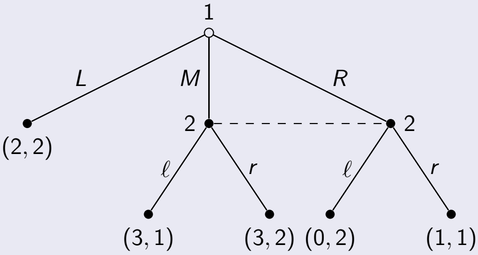
Definition

An assessment (β, μ) is ***sequentially rational*** if $\forall i \in \mathcal{N}$ and $\forall I_{i,j} \in \mathcal{I}_i$,

$$u_i\left((\beta, \mu) \,|\, I_{i,j}\right) \geq u_i\left(\left((\beta_i', \beta_{-i}), \mu\right) \,|\, I_{i,j}\right), \quad \forall \beta_i'$$

$u_i\left((\beta, \mu) \,|\, I_{i,j}\right)$ is i's utility from histories compatible with (β, μ) at $I_{i,j}$.

Example



$\beta = (M, r)$ is a NE (and also SPE, no proper subgame).
Information set $\{M, R\}$ is *on-equilibrium*: setting $\mu_2\left[\{M, R\}\right](M) = p$,

$$u_2(r) = 1 + p \geq u_2(\ell) = 2 - p \quad \Longrightarrow \quad p \geq 1/2.$$

The assessment (μ, β) with $p = 1$ is sequentially rational.

Beliefs, rationality and Bayes

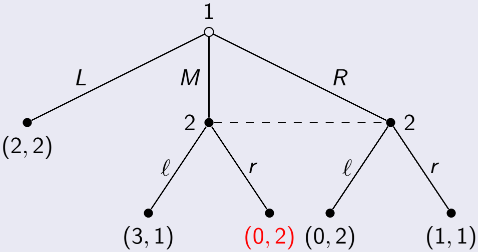
Proposition

A (behavioral) strategy profile β is a ***Nash equilibrium*** *of an extensive form game (of imperfect information) if and only if there exists a system of beliefs μ such that:*

1. *the assessment (β, μ) is* ***sequentially rational*** *at all information sets* ***on the equilibrium path****, i.e. $\forall I_{i,j}$ such that $\Pr[I_{i,j}|\beta] > 0$*
2. ***on the equilibrium path****, the belief system μ is derived from β through the* ***Bayes rule****, i.e. $\forall I_{i,j}$ such that $\Pr[I_{i,j}|\beta] > 0$, $\forall h \in I_{i,j}$ we have*

$$\begin{aligned} \mu_i[I_{i,j}](h) &= \Pr[h|I_{i,j}, \beta] = \frac{\Pr[I_{i,j}|h, \beta] \Pr[h|\beta]}{\Pr[I_{i,j}|\beta]} \\ &= \frac{\Pr[h|\beta]}{\Pr[I_{i,j}|\beta]}. \end{aligned}$$

Example



$\beta = (L, r)$ is a NE (and also SPE, no proper subgame).

The information set $\{M, R\}$ is now *off-equilibrium*.

For $p \geq 1/2$, action r is sequentially rational, but it is not necessary to check $\{M, R\}$ for (L, r) to be a NE.

Sequential Equilibrium

Zero probability events can be obtained as the limit of positive-probability events (so Bayes is always applicable).

Definition

The assessment (β, μ) is ***consistent*** if there is a sequence $\{(\beta^n, \mu^n)\}_{n=1}^{\infty}$ of assessments such that:

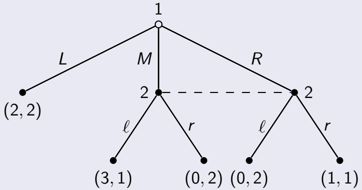
1. converges to (β, μ),
2. each β^n is fully mixed,
3. the belief system μ^n is derived from β^n using Bayes rule.

Definition

An assessment (β, μ) is a ***sequential equilibrium*** (SE) of a finite extensive-form game (of imperfect information) with perfect recall if it is sequentially rational and consistent.

Sequential Equilibrium

Example



The assessment (β, μ) with $\beta_1(L) = 1$, $\beta_2(r) = 1$, $\mu_2\left[\{M, R\}\right](M) = p$ is a **sequential equilibrium** for $p \geq 1/2$:

1) sequentially rational for $p \geq 1/2$ (s.t. $u_2(r) \geq u_2(\ell)$);

2) consistent, indeed $(\beta, \mu) = \lim_{\epsilon \to 0} (\beta^\epsilon, \mu^\epsilon)$ with

$$\beta_1^\epsilon = (1 - \epsilon, p\epsilon, (1-p)\epsilon), \text{ and } \beta_2^\epsilon = (\epsilon, 1 - \epsilon),$$

$$\mu_2^\epsilon\left[\{M, R\}\right](M) = \frac{p\epsilon}{\epsilon} = p, \text{ and } \mu_2^\epsilon\left[\{M, R\}\right](R) = \frac{\epsilon(1-p)}{\epsilon} = 1 - p.$$